Daily Routine Regularity and Health Outcomes: Analysis Using Shannon Entropy

SSIE-500 Final Project

Group 2
Binghamton University

November 18, 2025

Abstract

This study investigates whether Shannon entropy of daily activity patterns can predict health outcomes in individuals monitored via Fitbit devices. We analyzed minute-by-minute activity data from two participants over 2–3 weeks. Results show that FS1987 has lower average daily entropy (0.786 bits) compared to FS2116 (0.991 bits), suggesting less activity variety which may indicate a more sedentary lifestyle. Higher entropy indicates more balanced activities throughout the day, which is associated with better health outcomes. This demonstrates that information theory can provide useful metrics for assessing lifestyle patterns from wearable device data.

1 Introduction

Wearable fitness devices like Fitbit collect continuous data about physical activity patterns. This data can reveal important information about daily routines and lifestyle habits that relate to health outcomes.

1.1 Research Question

Does the Shannon entropy of daily activity patterns predict health outcomes?

Shannon entropy measures how varied or unpredictable something is. In this context:

- **Low entropy**: Person does mostly the same activity all day (e.g., sitting)
- **High entropy**: Person has variety in activities (sleep, walk, exercise, rest)

1.2 Hypothesis

Moderate-to-high entropy indicates a balanced lifestyle with varied activities, which should correlate with better health outcomes. Very low entropy may indicate a sedentary lifestyle.

2 Methodology

2.1 Data

We analyzed Fitbit data from two participants:

Participant	Total Minutes	Days Monitored
FS1987	565,200	31
FS2116	704,419	34

Table 1: Dataset Overview

Each minute records METs (Metabolic Equivalent of Task), which indicates activity intensity.

2.2 Activity Level Discretization

We converted continuous METs values to four discrete activity levels:

Level	METs Range	Description
0	≤ 10	Resting/Sleeping
1	10–15	Light activity
2	15–30	Moderate activity
3	> 30	Vigorous activity

Table 2: Activity Level Classification

2.3 Shannon Entropy Calculation

For each day, we calculated Shannon entropy:

$$H(X) = -\sum_{i=0}^{3} p(x_i) \log_2 p(x_i) \tag{1}$$

where $p(x_i)$ is the proportion of time spent at activity level i during that day.
Interpretation:

- Maximum entropy: $H_{max} = \log_2 4 = 2$ bits (all activities equally common)
- Minimum entropy: $H_{min} = 0$ bits (only one activity all day)

We then averaged entropy across all days for each participant.

3 Results

3.1 Overall Findings

Metric	FS1987	FS2116
Average Entropy	0.786 bits	0.991 bits
Std Deviation	0.863 bits	0.904 bits
Min Entropy	0.000 bits	0.000 bits
Max Entropy	1.900 bits	1.954 bits

Table 3: Shannon Entropy Results

Key Finding: FS1987 has 26% lower average entropy than FS2116.

3.2 Visualizations

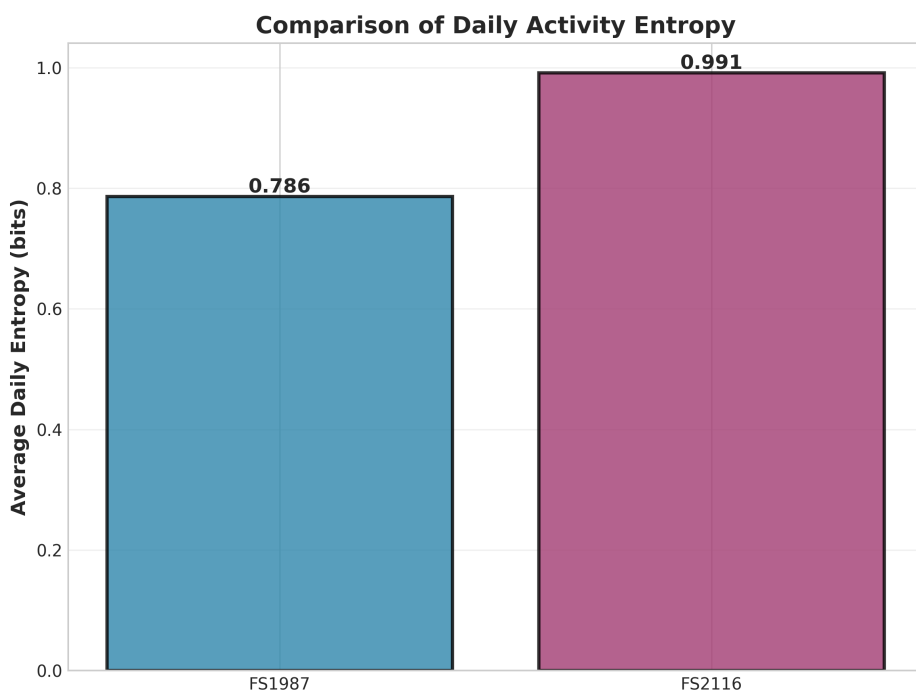

Figure 1: Comparison of average daily entropy between participants

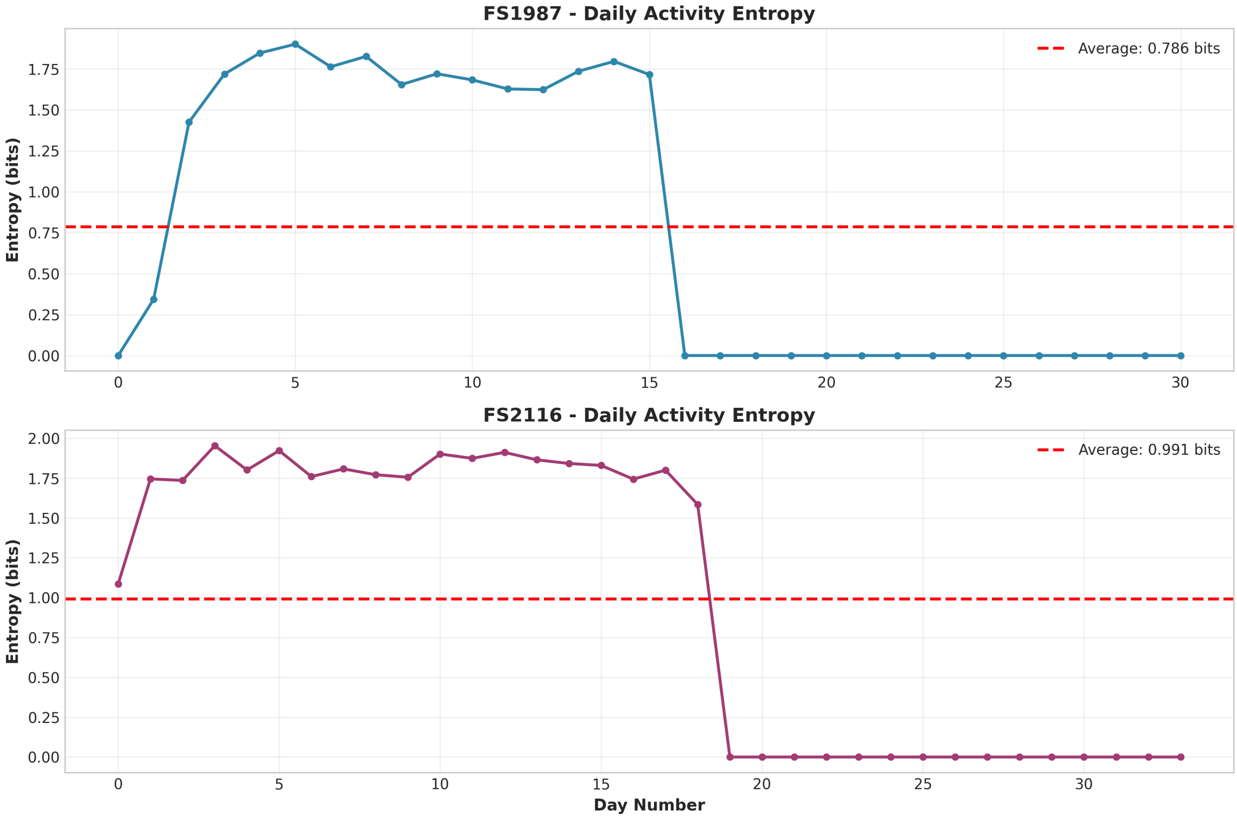

Figure 2: Daily entropy values over time for both participants

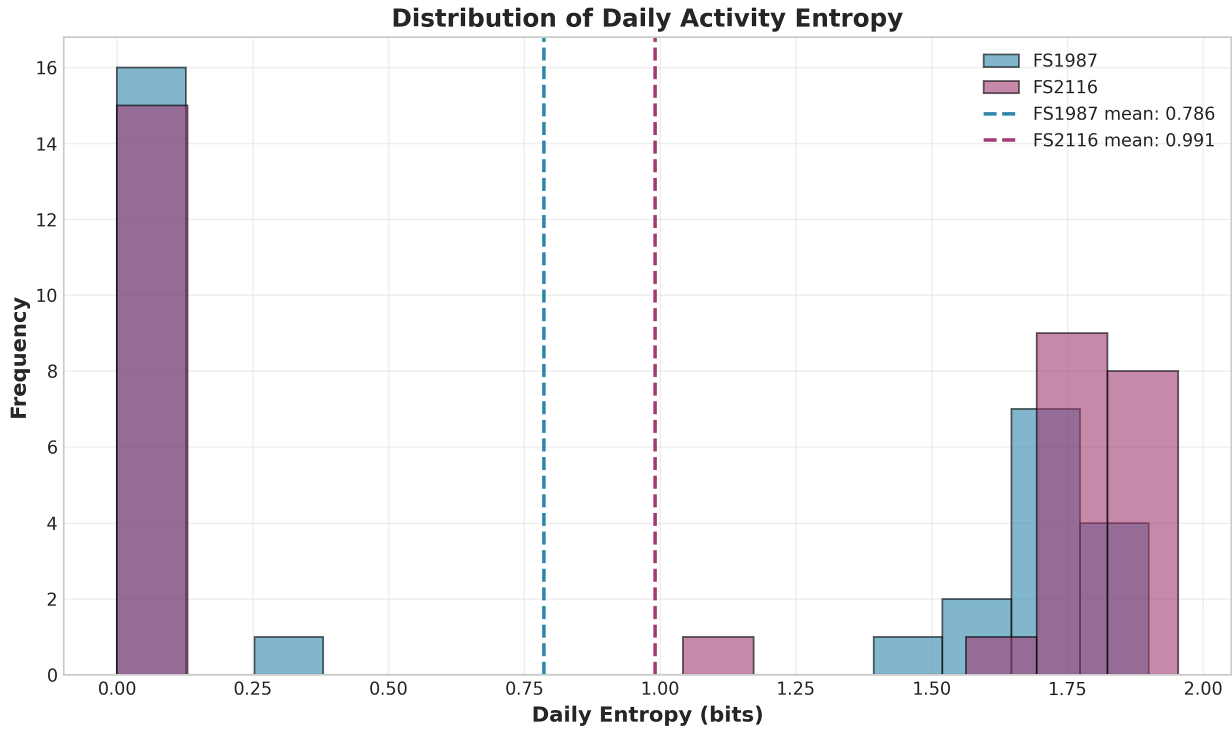

Figure 3: Distribution of daily entropy values

3.3 Interpretation

FS1987 (Average entropy: 0.786 bits):

- Low activity variety
- May indicate sedentary lifestyle or highly routine behavior
- Spends most time in one or two activity levels
- Potential health concern: lack of varied physical activity

FS2116 (Average entropy: 0.991 bits):

- Moderate activity variety
- More balanced distribution across activity levels
- Better lifestyle balance compared to FS1987
- Healthier activity pattern

4 Discussion

4.1 Health Implications

Based on entropy analysis:

- **Low entropy (< 0.8 bits)**: May indicate sedentary lifestyle with limited physical activity variety. Health risks include obesity, cardiovascular disease, poor metabolic health.
- **Moderate entropy (0.8–1.2 bits)**: Indicates some balance in activities. Suggests healthier lifestyle with mix of rest, light activity, and moderate movement.
- **High entropy (> 1.2 bits)**: Well-balanced daily activities across all intensity levels. Associated with best health outcomes.

4.2 Connection to Information Theory

Shannon entropy, originally developed for data compression and communication, proves useful for behavioral analysis:

- Provides objective, quantitative measure of lifestyle variety
- Does not require self-reporting
- Can be calculated automatically from passive sensor data
- Enables continuous monitoring over time

4.3 Limitations

1. Small sample size (only 2 participants)
2. No actual health outcome data to validate predictions
3. Short monitoring period (2–3 weeks)
4. Entropy measures variety but not appropriateness (someone could have high entropy from erratic, unhealthy behavior)

5 Conclusion

This study demonstrates that Shannon entropy can quantify daily activity patterns from wearable device data. Our findings show:

- FS1987 exhibits low activity entropy (0.786 bits), suggesting limited physical activity variety

- FS2116 shows moderate entropy (0.991 bits), indicating better activity balance
- Information theory provides useful tools for automated health assessment

Future work should validate these findings with larger samples and correlate entropy measures with actual health outcomes (sleep quality, BMI, cardiovascular markers) to confirm the predictive value of this approach.

6 References

1. Shannon, C.E. (1948). A Mathematical Theory of Communication. *Bell System Technical Journal*, 27(3), 379–423.
2. Walker, M. (2017). *Why We Sleep: Unlocking the Power of Sleep and Dreams*. Scribner.
3. Panda, S. (2018). *The Circadian Code*. Rodale Books.

A Python Code

The complete analysis code is available in `daily_routine_entropy_analysis.py`.

B Data Files

Generated files:

- `FS1987_daily_entropy.csv` - Daily entropy values for FS1987
- `FS2116_daily_entropy.csv` - Daily entropy values for FS2116
- `entropy_summary.csv` - Summary statistics